DESCENT MORPHISMS, EFFECTIVE DESCENT MORPHISMS AND UNIVERSAL INJECTIVITY FOR MODULES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement is terse – it speaks of 'the morphism f' and of the two descent properties for modules without recalling them – so two author definition environments are delivered as context spans: the definition of descent data for modules relative to a ring map, which fixes f and the category of descent data, and the definition naming the two descent properties. Both are the author's own text, carried verbatim, and neither is counted as a statement of the card.

Editorial note. The theorem is the one the later parts of the chapter invoke as 'Theorem theorem-descent'; the card reproduces exactly the counted span and the two context spans.

Editorial note. The counted spans carry 3 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Descent data for modules relative to a ring map and the base extension functor. Same chapter, Definition definition-descent-datum-modules, delivered as the context span at lines 487-523. Fixes the ring map f, the category of descent data and the functor whose full faithfulness and essential surjectivity are at issue.
- Descent morphism for modules and effective descent morphism for modules. Same chapter, Definition definition-effective-descent, delivered as the context span at lines 1314-1321. Supplies the two properties named in parts (a) and (b) of the statement.
- The natural map from the base extension of the pushforward of a descent datum back to the datum is an isomorphism. Same chapter, Lemma lemma-equalizer-M and Lemma lemma-descent-lemma, invoked at proof lines 1525-1528. Provides the isomorphism of the counit which the adjunction argument consumes.
- Base change along a universally injective ring map reflects isomorphisms for the unit of the adjunction. Algebra, Lemma algebra-lemma-base-change-along-universally-injective-ring-map, invoked at proof lines 1533-1535. Upgrades the fact that the unit becomes an isomorphism after base extension to the fact that the unit itself is an isomorphism.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0H9I: sources/stacks-chow/tags.txt, source lines 21528–21528. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 08X7: sources/stacks-descent/pinned-original-source.tex, source lines 1430–1430. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 08WZ: sources/stacks-descent/pinned-original-source.tex, source lines 1329–1329. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let $R \to A$ be a ring map.

(1) A *descent datum* (N, φ) *for modules with respect to* $R \to A$ is given by an A-module N and an isomorphism of $A \otimes_R A$-modules

$$\varphi : N \otimes_R A \to A \otimes_R N$$

such that the *cocycle condition* holds: the diagram of $A \otimes_R A \otimes_R A$-module maps

$$\begin{array}{ccc} N \otimes_R A \otimes_R A & \xrightarrow{\varphi_{02}} & A \otimes_R A \otimes_R N \\ & {\scriptstyle \varphi_{01}} \searrow \quad \nearrow {\scriptstyle \varphi_{12}} & \\ & A \otimes_R N \otimes_R A & \end{array}$$

commutes (see below for notation).

(2) A *morphism* $(N, \varphi) \to (N', \varphi')$ *of descent data* is a morphism of A-modules $\psi : N \to N'$ such that the diagram

$$\begin{array}{ccc} N \otimes_R A & \xrightarrow{\varphi} & A \otimes_R N \\ {\scriptstyle \psi \otimes \mathrm{id}_A} \downarrow & & \downarrow {\scriptstyle \mathrm{id}_A \otimes \psi} \\ N' \otimes_R A & \xrightarrow{\varphi'} & A \otimes_R N' \end{array}$$

is commutative.

Definition 0.2. The functor $f^* : \mathrm{Mod}_R \to DD_{S/R}$ is called *base extension along* f. We say that f is a *descent morphism for modules* if f^* is fully faithful. We say that f is an *effective descent morphism for modules* if f^* is an equivalence of categories.

Theorem 0.3. *The following conditions are equivalent.*

(a) *The morphism f is a descent morphism for modules.*
(b) *The morphism f is an effective descent morphism for modules.*
(c) *The morphism f is universally injective.*

Proof. It is clear that (b) implies (a). We now check that (a) implies (c). If f is not universally injective, we can find $M \in \mathrm{Mod}_R$ such that the map $1_M \otimes f : M \to M \otimes_R S$ has nontrivial kernel N. The natural projection $M \to M/N$ is not an isomorphism, but its image in $DD_{S/R}$ is an isomorphism. Hence f^* is not fully faithful.

We finally check that (c) implies (b). By Lemmas 08WZ and 08X7, for $(M, \theta) \in DD_{S/R}$, the natural map $f^*f_*(M, \theta) \to M$ is an isomorphism of S-modules. For $M_0 \in \mathrm{Mod}_R$ we want to show $M_0 \to f_*f^*M_0$ is an isomorphism. By adjunction, the composition $f^*M_0 \to f^*f_*f^*M_0 \to f^*M_0$ is the identity, while the above shows that the second arrow is an isomorphism. Thus $f^*M_0 \to f^*f_*f^*M_0$ is an isomorphism. By Algebra, Lemma 0H9I we conclude $M_0 \to f_*f^*M_0$ is an isomorphism. Consequently, f_* and f^* are quasi-inverse functors, proving the claim. □